# AAE 351: Aerospace Systems Design – Spring 2026 Syllabus
# 3 credits

## Course Information/Description

**Official course description in the university catalog:** Aerospace system lifecycle and design process. Stakeholder needs elicitation and requirements generation. Quality function deployment and hierarchical objective trees. Concept generation and creativity techniques. Introduction to safety, risk, cost and value analysis. Critical evaluation of the applicability of systems engineering techniques in specific contexts. Application of these techniques to a team semester design project.

## Learning Resources & Reference Texts

Access the course videos, materials, and resources via Purdue's Brightspace learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

**Reading and Other Learning Resources:** All required reading materials, media, online assessments, and other learning resources will either be provided in Brightspace, handed out in class, or a link will be provided so students may access these resources on their own.

## Course Learning Outcomes

The learning outcomes of this course are:

1. Identify appropriate tools, methods, and processes to formulate an aerospace system design problem including realistic constraints from technical, economic, social, political, safety, and other relevant contexts
2. Apply those tools, methods, and processes to generate solutions to the aerospace system design problem

## Assignments

Your achievement of course learning outcomes will be assessed through a combination of participation, homework assignments, project briefings, and feedback opportunities spread throughout the academic period. Details on these assignments, including a schedule of due dates, rubrics to guide evaluation, and guidelines for how to participate and give effective feedback will be posted on the course website and discussed in class. Generally, homework will be due on Mondays at 2pm, but there will be exceptions so visit Brightspace often to confirm.

The detailed course schedule, including dates for assignments, project briefings, and lectures is available on Brightspace.

## Evaluation/Grading

**Grading Philosophy:** One of my primary aims in this course is to model the aerospace industry wherever possible, so students understand what the environment is like and how to succeed within that environment. The equivalent of a 'course grade' in industry is usually an 'employee annual assessment.' The key factors used in these assessments can vary greatly, but some form of the following foundational elements is nearly always included:

- The overall performance that year of your company, business unit, and the specific program/project/team you worked on
- Direct customer feedback on your team's (and sometimes your individual) performance
- How well your behaviors align with the culture/values of the group you are in
- Your reputation for how well you work with others (e.g. Are you a good team player? Are you easy/hard to work with? Did you make your team and other teams better by proactively helping others?)
- How important/valuable your job/role was to the success of the overall project(s)
- How well you delivered on your commitments for that job/role

These factors stand in stark contrast to usual university grading schemes because industry evaluations 1) are largely subjective, 2) don't grade on a curve, and 3) are holistic in nature (not just what you did, but how you worked). My goal is to align the grades of this class as much as possible to factors such as these while preserving the required element of assigning a letter grade to each student at the end of the semester.

**What does this mean for this course?** While specific grading rubrics will be forthcoming relative to team projects throughout the semester, it means that for this course:

- Your performance in this class will be evaluated holistically
- You will be graded both on your personal work as well as the work of your team
- You will be graded on your efforts to help other teammates and other teams be successful. In other words, cooperation and collaboration will be rewarded, **not competition**.
- There will not be a forced distribution of grades at the end

With the bullets above in mind, here is a general framework for relative contribution of different elements of this course to the final letter grade.

| Graded Item | Contribution to Course Grade |
|---|---|
| Personality & Teaming Homework | 10% |
| Project Phase 1 Presentation & Written Reflection | 40% |
| Project Phase 2 Presentation & Written Reflection | 50% |
| Contributions to Other Team's Success | Part of Phase 1 & 2 Grade |